

Java Assignment 03: Loops, Switch & Array Processing

Java Assignment – 3

1. Reverse a Number

Write a Java program to read an integer in the main() method and reverse its digits. Use a static method to perform the required logic. Pass the integer as a parameter to the method and return the reversed number to the main() method. Display the original number and the reversed number.

2. Sum of Digits

Write a Java program to read an integer in the main() method and find the sum of its digits. Use a static method to perform the required logic. Pass the integer as a parameter to the method and return the sum to the main() method. Display the calculated sum.

3. Multiplication Table

Write a Java program to read a number in the main() method and generate its multiplication table from 1 to 10. Use a static method to perform the required logic. Pass the number as a parameter to the method and return the result to the main() method.

4. Prime Numbers

Write a Java program to read a number n in the main() method and find all prime numbers between 1 and n. Use a static method to perform the required logic. Pass n as a parameter to the method and return the result to the main() method. Display all the prime numbers.

5. Simple Calculator

Write a Java program to read two numbers and an arithmetic operator (+, -, *, /, %) in the main() method. Use a static method to perform the selected operation. Pass the numbers and operator as parameters to the method and return the calculated result to the main() method. Display the result.

6. Menu-Driven Area Calculator

Write a Java program to display the following menu and read the user's choice in the main() method:

Area of Circle

Area of Rectangle

Area of Triangle

Exit

Read the required values in the main() method. Use a static method to perform the area calculation based on the selected choice. Pass the required values and choice as parameters to the method and return the calculated area to the main() method. Display the result.

7. Find Maximum and Minimum

Write a Java program to read n integers into an array in the main() method and find the maximum and minimum elements. Use a static method to perform the required logic. Pass the array as a parameter to the method and return the maximum and minimum results to the main() method. Display both values.

8. Student Marks Analysis

Write a Java program to read the marks of n students into an array in the main() method. Use a static method to calculate the total, average, highest mark, and lowest mark. Pass the array as a parameter to the method and return the required results to the main() method. Display the results.

9. Search an Element

Write a Java program to read n integers into an array and a search element in the main() method. Use a static method to search for the given element in the array. Pass the array and search element as parameters to the method and return the position of the element to the main() method. Display whether the element is present and its position if found.

10. Count Even and Odd Numbers

Write a Java program to read n integers into an array in the main() method and count the number of even and odd elements. Use a static method to perform the required logic. Pass the array as a parameter to the method and return the even and odd counts to the main() method. Display both counts.